

MUTO

Two variable-mu valve compressors in one: a 1960s-style vintage limiter and a modern stereo vari-mu, each with its own faceplate and settings.



Overview

MUTO is a pair of variable-mu valve compressors. The VINTAGE and MODERN keys in the header swap both the faceplate and the circuit, and each unit keeps its own settings, so you can flip between them and compare.

Both work the same way at heart. The gain element is a remote-cutoff valve whose gain falls smoothly as its bias rises, and the bias comes from a rectifier on the unit's own output (feedback detection). The harder it works, the steeper it gets: the ratio rises with the gain reduction, giving a soft, ever-tightening knee. The valve also colours the sound more the deeper it is biased, a little of the control voltage leaks through as a soft thump when the bias moves fast, and the output iron thickens the lows.

MU-1 VINTAGE is in the spirit of a famous London studio's 1960s in-house rebuild of a broadcast valve limiter: simple controls, a fixed fast attack (about 2 ms), six RECOVERY positions and a HOLD mode. It is about 2:1 at light settings, rising to around 8:1 at 10 dB of gain reduction. MU-2 MODERN is in the spirit of today's boutique all-valve vari-mu units: adjustable attack and release, side-chain high-pass filters, stereo, dual-mono and mid-side modes, and a heavier, adjustable output transformer. Its gain law is gentler (about 4:1 at 6 dB, 7:1 at 15 dB) and its valve stage cleaner.

Reach for it for

- Smooth, musical levelling of vocals, bass and piano
- Gluing a mix or drum bus with a few dB of gentle gain reduction
- Thick, dense, pumping room and drum sounds (VINTAGE with HOLD)
- Mid-side bus compression that leaves the width alone (MODERN in M-S)
- Parallel compression and low-end weight with MIX and IRON
- Mastering touches with slow attack and AUTO release

Controls

The header

VINTAGE / MODERN	Chooses the unit: MU-1 VINTAGE or MU-2 MODERN. The faceplate and the circuit change together; each unit remembers its own settings.
PRESETS	Opens the full preset list.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset number and name. Click it for the preset list too.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The meters (both units)

Two needle meters sit in the middle of the panel. In GR mode the needle rests at 0 and swings left by the gain reduction in dB. In OUT mode they read the output level with VU ballistics, calibrated so that a sine peaking at -18 dBFS reads 0 VU. The caption under each meter shows what it is following: LINKED, LEFT / RIGHT, or MID / SIDE in the modern unit's M-S mode (OUT always reads LEFT and RIGHT).

MU-1 VINTAGE

A blue-grey enamel faceplate with black knobs and white legends.

INPUT	A stepped attenuator, -8 to +24 dB in 2 dB steps (0 dB by default). It drives the valves: more input gives more gain reduction and more colour.
THRESHOLD	The side-chain sensitivity, -40 to 0 dB (-10 dB by default). Lower settings compress more.
RECOVERY	The release, in six steps (3 by default): 1 = 80 ms, 2 = 200 ms, 3 = 450 ms, 4 = 1 s (3 and 4 with a little memory), 5 = two-stage 150 ms / 2.5 s, 6 = two-stage 300 ms / 6 s. The two-stage settings are programme dependent: short peaks recover fast, but sustained compression charges a second, slow store and lets go slowly.
OUTPUT	Make-up gain, -20 to +20 dB.
STEREO: LINK / DUAL	LINK (the default) ties the two channels together so the stereo image stays put; DUAL lets each side compress on its own, like two mono units.
METER: GR / OUT	What the needles show (see above).
MIX	Parallel compression: blends the compressed signal with the dry one, 0 to 100% (100% by default). The dry signal is level-matched (it goes through INPUT and OUTPUT too).
DRIFT	Valve ageing, 0 to 100% (50% by default). See Drift below.
OFF / CLIP / LIMIT (OUT)	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency, the default), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). While the output clips, the CLIP key flashes red.
RECOVERY MODE: NORMAL / HOLD	HOLD drives the side-chain hard until it saturates into a long, slow hold (1.6 s / 12 s): the valves sit deep in their bias, so the sound gets thick and dense, with more distortion and more thump. While HOLD is on, RECOVERY has no effect.
POWER	The pilot lamp.

MU-2 MODERN

A deep anodised faceplate with machined aluminium knobs and warm-lit meters.

INPUT	Drive into the valves, -12 to +24 dB (0 dB by default).
THRESHOLD	-40 to 0 dB (-12 dB by default).
ATTACK	0.1 to 30 ms (10 ms by default), from snappy to valve-sluggish. Slow settings let the transients through before the compressor grabs.
RELEASE	50 ms to 3 s (300 ms by default). Ignored while AUTO is on (the value reads AUTO).
HPF L / HPF R	A high-pass filter on each side-chain (not on the audio): OFF, 60, 90, 150 or 250 Hz. It stops the low end from driving the compression, so bass and kick don't pump the whole mix. In M-S mode the two read HPF M and HPF S.
MODE: STEREO / DUAL MONO / M-S	STEREO: left and right, linked by LINK. DUAL MONO: two independent channels. M-S: the valves work on the mid (centre) and the side (width) instead, and the result is turned back into left and right.
LINK	How tightly the two side-chains are tied, 0 to 100% (100% by default), in STEREO and M-S. Not used in DUAL MONO.
IRON	The weight of the output transformer, 0 to 100% (30% by default): more low-end saturation and a gentle low shelf.
MIX	Parallel compression, 0 to 100% (100% by default), against a level-matched dry signal.
OFF / CLIP / LIMIT (OUT)	The final stage, as on the vintage unit.
AUTO (RELEASE)	A two-stage, programme-dependent release: fast after short peaks, slow after sustained compression.
GR / OUT (METER)	Switches the meters between gain reduction and output level; the key shows the current choice.

DRIFT	Valve ageing, 0 to 100% (50% by default).
OUTPUT	Make-up gain, -20 to +20 dB.

Drift

DRIFT is valve ageing. At 50% the bias and threshold wander slowly by about 0.4 dB, the two channels' gain is mismatched by about 0.1 dB (a pair of valves never match exactly), and the recovery times wander a little. On the vintage unit a faint 50 Hz mains hum (about -90 dBFS at 50%, a touch more with HOLD) wanders with it. At 0% both channels match and stay still.

Knobs: drag up or down (Shift for fine moves) or use the mouse wheel; double-click to reset a knob to its default. A knob's value shows while you touch it.

Presets

Each preset switches to its own unit and sets that unit's controls; the other unit's settings are left as they are. DRIFT, the output stage and the meter mode are left alone, except by Init.

Init	Every control on both units back to its default, on VINTAGE, with DRIFT 50%, CLIP and the GR meter.
Vintage Vocal	+4 dB in, threshold -14 dB, RECOVERY 3, +3 dB out: smooth vocal levelling.
Vintage Drums Hold	+4 dB in, RECOVERY 4 with HOLD, 80% MIX: dense, thick drums with some dry punch kept.
Vintage Bass	+6 dB in, RECOVERY 2 for a fast, even bass line.
Room Hold Crush	+12 dB in, threshold -20 dB, RECOVERY 6 and HOLD, +7 dB out: heavy, pumping room mics.
Piano Leveller	+2 dB in and the two-stage RECOVERY 5: gentle, programme-dependent levelling.
Modern Mix Bus	Attack 20 ms, AUTO release, 60 Hz side-chain filters, IRON 40%: bus glue.

Modern Drum Bus Snap	Fast 1 ms attack, 120 ms release, 90 Hz side-chain filters, IRON 50%, +3 dB out.
Modern M-S Glue	M-S mode with LINK 20%, AUTO release, 60 Hz on the mid and 150 Hz on the side.
Modern Vocal	+4 dB in, 3 ms attack, 250 ms release, 150 Hz side-chain filters.
Modern Mastering Touch	Threshold -12 dB, slow 30 ms attack, AUTO release, IRON 25%, +0.5 dB out.
Modern Parallel Weight	+6 dB into a -24 dB threshold, IRON 80%, MIX 40%: parallel weight and low end.

Tips

- Vari-mu compression works best with modest gain reduction: aim for 2 - 4 dB on a bus and let the rising ratio catch the peaks.
- On the vintage unit, INPUT and THRESHOLD both push it harder, but INPUT also adds colour - raise INPUT for character, lower THRESHOLD for control.
- Use the side-chain HPF at 90 or 150 Hz on a mix bus so the kick doesn't duck everything else.
- M-S with a lower LINK compresses the centre (vocal, kick, bass) without pulling in the width.
- Pull MIX back to 30 - 60% with heavy settings (or HOLD) for parallel compression that keeps the transients.
- Set DRIFT to 0 for a perfectly matched, repeatable pair.

File: gfx-muto.jsfx